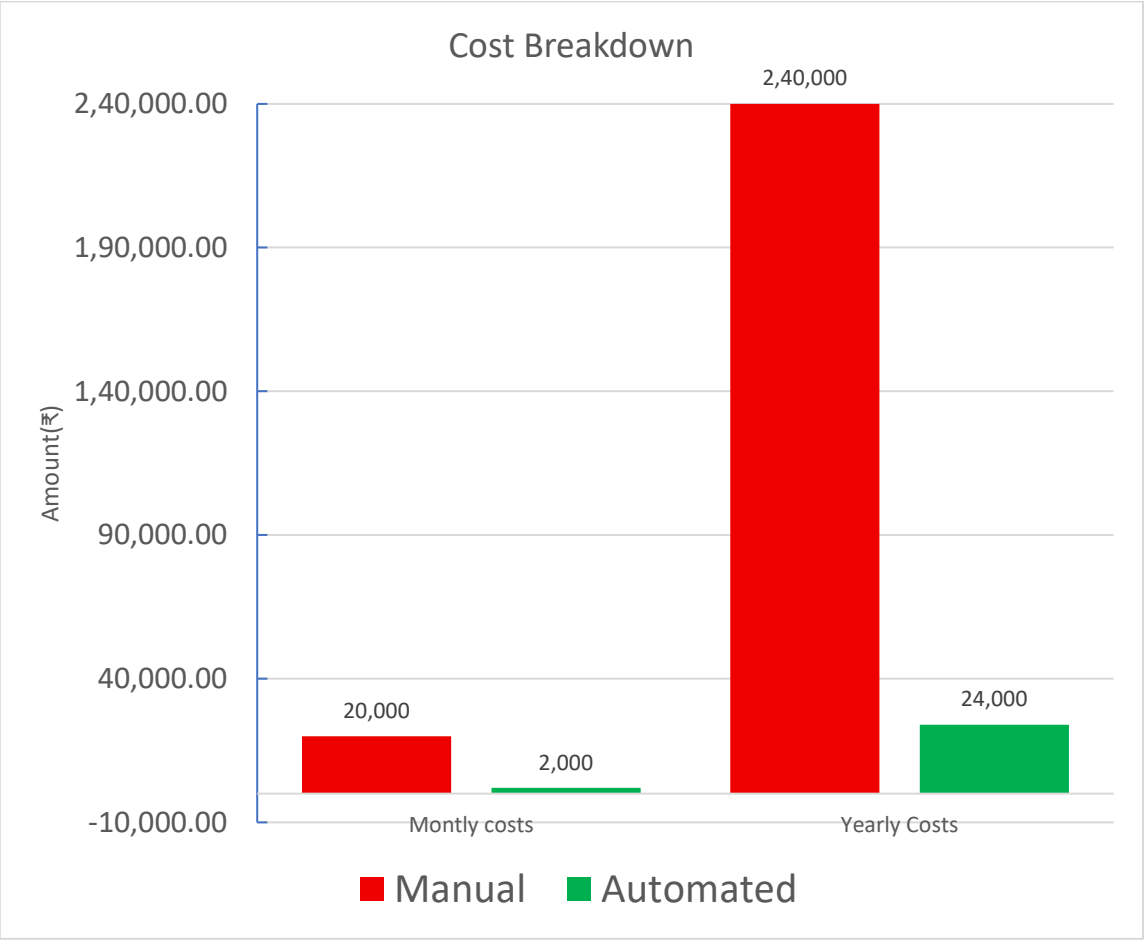
Analysis of Productivity in Automated Inspection

Estimated cost saving analysis

- **One manual inspector costs around ₹20,000.**
- **While the system is one time investment, it does requires electricity and maintenance cost. The monthly cost for electricity and maintenance is ₹2,000 and yearly cost is ₹24,000.**



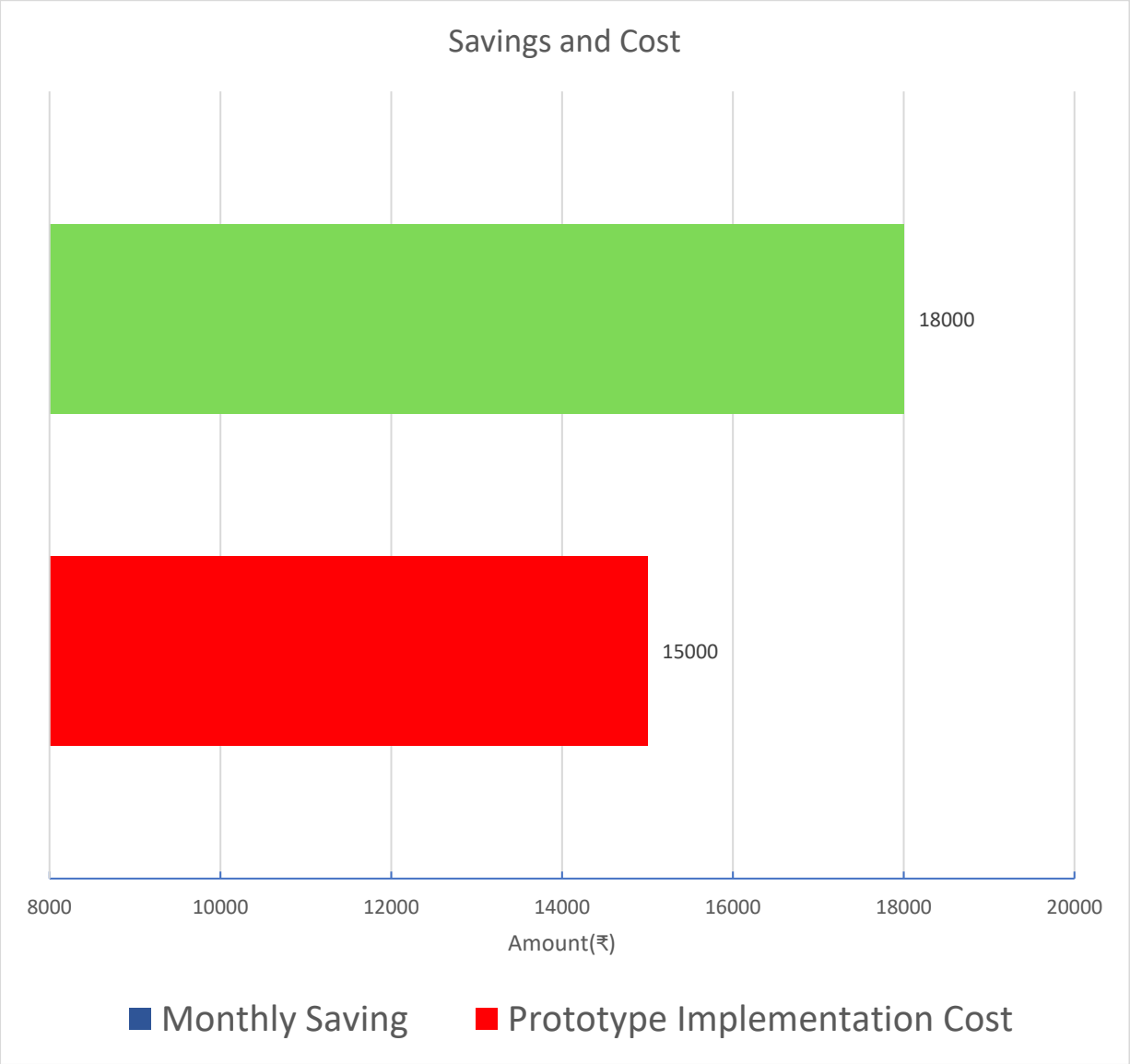
If automation reduces one manual inspector operator, then

Monthly Savings=₹18,000

Annual Savings=₹2,16,000

Return on Investment

- **The prototype implementation cost is ₹ 15,000.**
- **The monthly savings is ₹ 18,000 so investment recovery happens in about 25 days.**



Manual human inspection accuracy is approximately 85% to 90%.

While the YOLO v8n model which was trained on a custom dataset of approximately 1000 images has accuracy of defect detection of 94.6%.

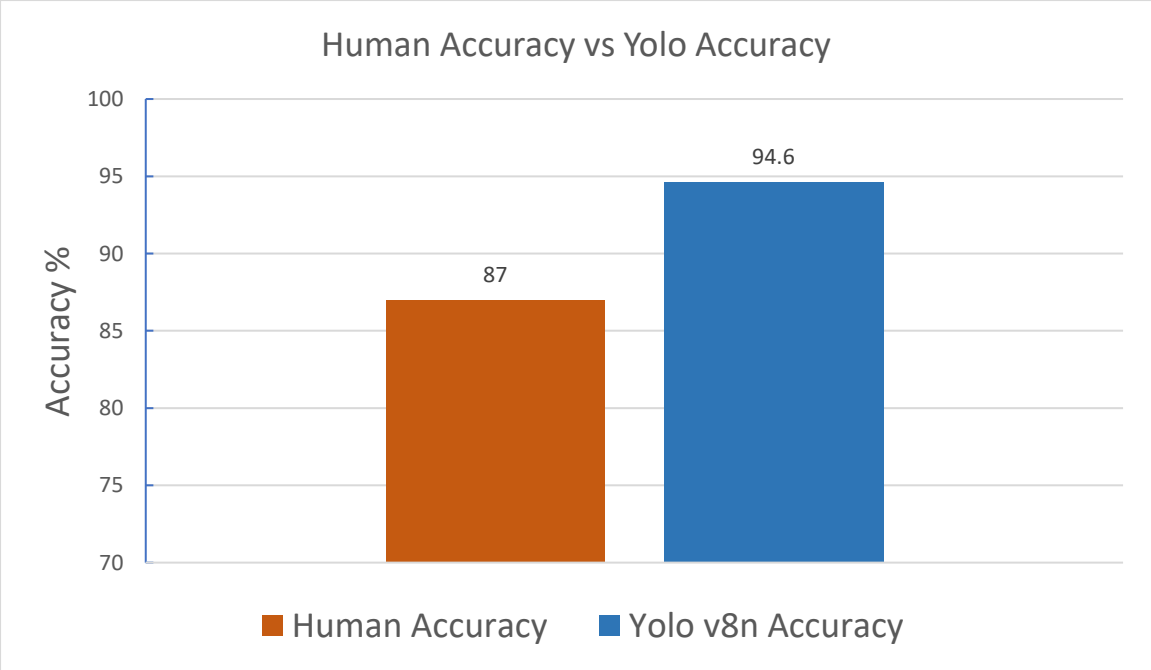
Yolo v8n model has 3.2 million parameters

Defect detection:

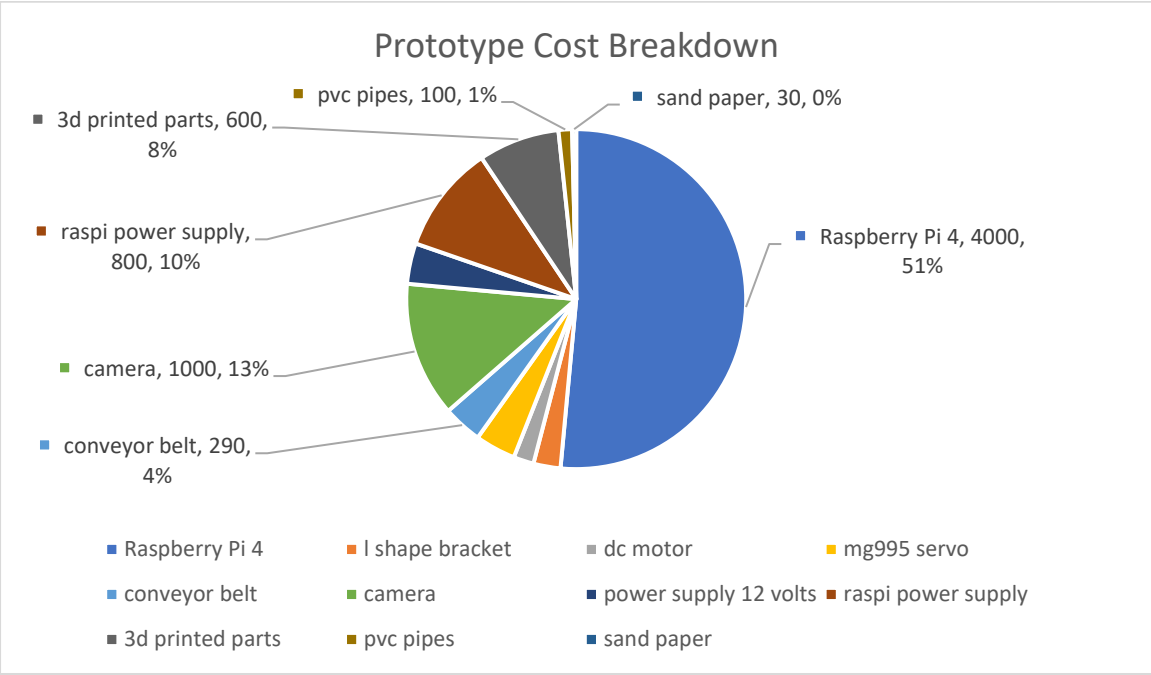
- **Precision = 87.2%**
- **Recall = 96%**
- **mAP@50 = 94.6%**

Gear detection:

- **Precision = 97.8%**
- **Recall = 100%**
- **mAP@50 = 99.5%**

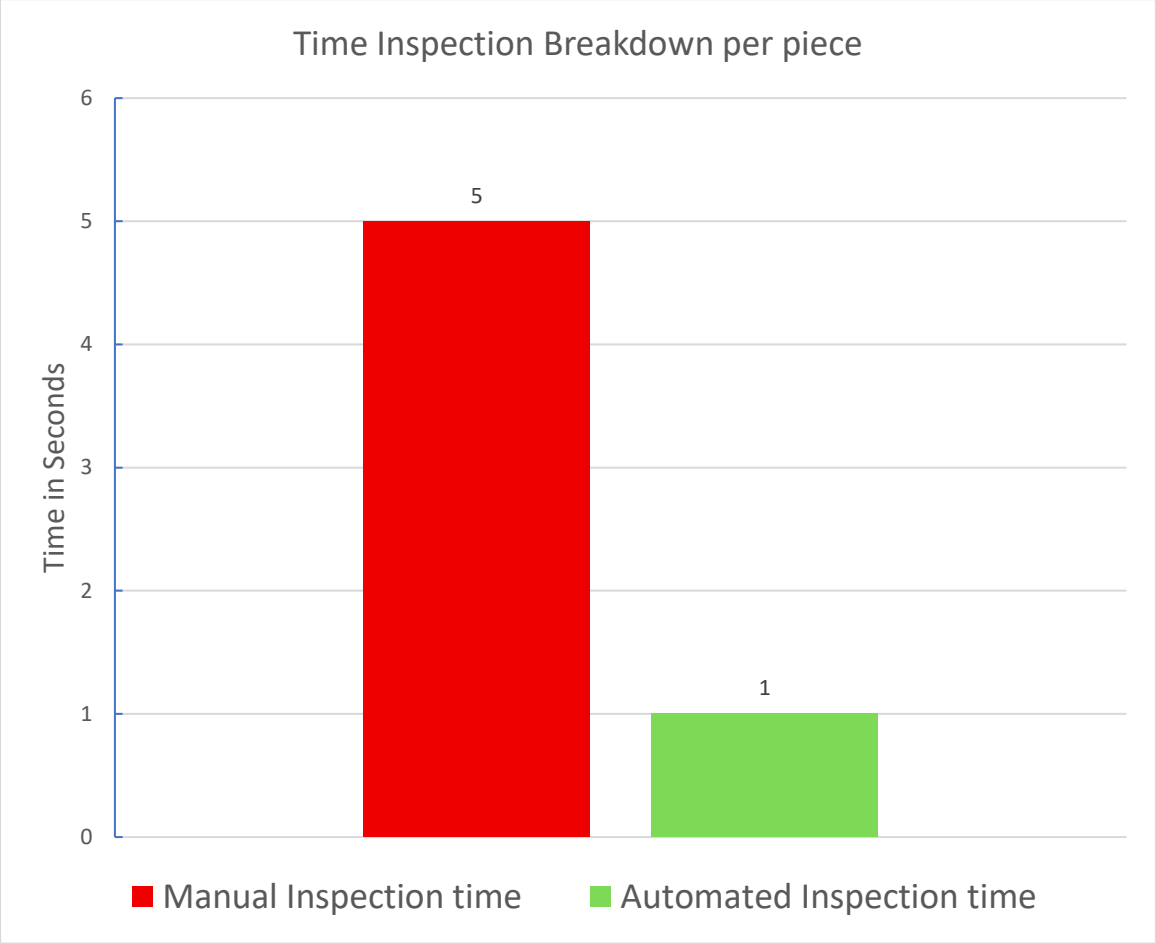


Total Cost for the Prototype is ₹ 8770



Raspberry Pi 4	4000
L shape bracket	200
DC motor	150
mg995 servo	300
conveyor belt	290
camera	1000
power supply 12 volts	300
raspi power supply	800
3d printed parts	600
pvc pipes	100
sand paper	30

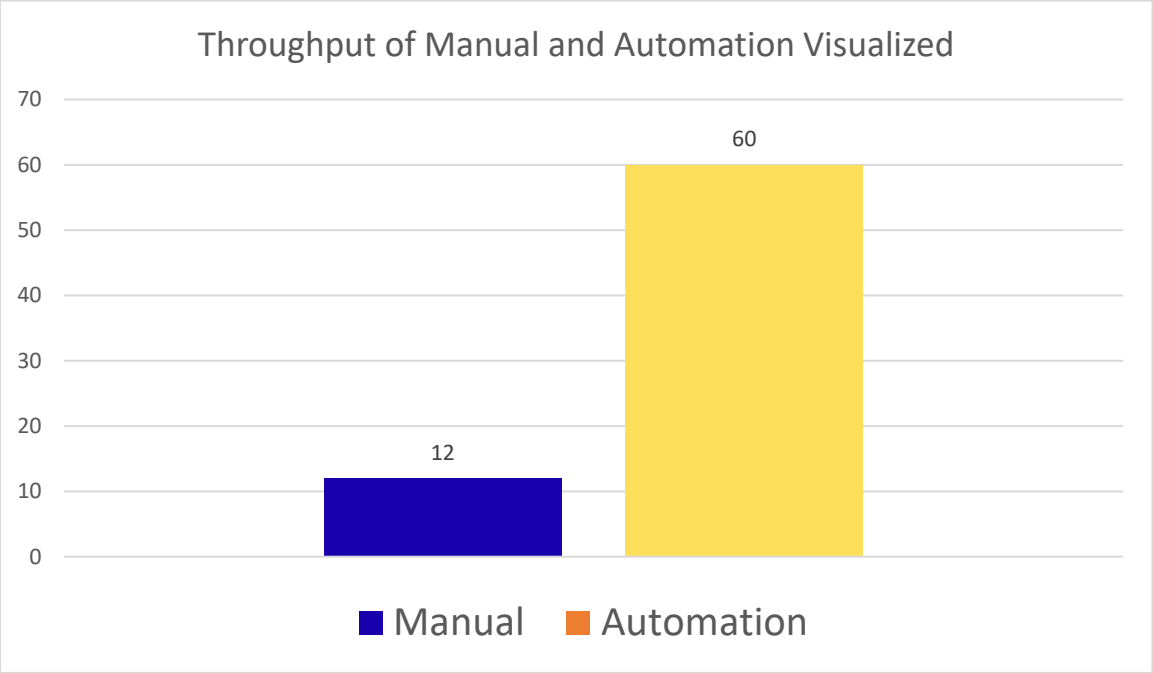
- **On Average, manual inspection for defects takes 5 seconds while automated inspection using YOLO takes 1 second.**



- **This shows that inspection time is reduced by 5x which saves time in the manufacturing process thus increasing the productivity.**
- **The time taken by the rejection mechanism to push out defected objects is 2 seconds.**

Throughput

- **Manual checks require 5 seconds per gear**
- **Manual=12 gears/min**
- **Automation=60 gears/min**



- **Therefore the throughput increases nearly 4x to 5x**